

1MT495 Current Issues in Applied International Economics: Reserve Currencies

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November 26, 2025

Outline

- 1 Reserve Currencies: Definitions and History
- 2 Current Landscape and Strategic Implications
- 3 Optimum Currency Areas (OCA)
- 4 Case Studies and Conclusion

Reserve Currencies: The Basics

- **Definition:** Foreign currencies held by central banks for international transactions, stability, and reserves.
- **Dominance:** Driven by the issuer's economic size, stability, and **network effects** (inertia).
- **Key Characteristics:**
 - **Liquidity:** Deep, accessible financial markets.
 - **Stability:** Low inflation/default risk.
 - **Convertibility:** Unrestricted capital flows.
 - **Institutional Strength:** Rule of law, central bank independence.

Taxonomy of International Money

Cohen's (1971) Functional Mapping

Distinguishing between private and official roles is essential.

Function	Private Use	Official Use
Unit of Account	Trade Invoicing	Pegging Anchor
Medium of Exchange	Vehicle Currency	Intervention Currency
Store of Value	Investment Assets	Reserve Assets

Table: The Roles of an International Currency

Importance of Reserve Currencies: The Benefits

- **Facilitating Global Trade**

- Reduces currency conversion costs and exchange rate risks.
- Most commodities (e.g., oil, gold) are priced in the reserve currency (USD).

- **Store of Value and Liquidity**

- Central banks hold reserves for market interventions and crisis buffers (e.g., during COVID-19).

- **Economic Influence for Issuing Country**

- "Exorbitant privilege": Lower interest rates, seigniorage revenue, and geopolitical leverage.
- Example: The US can borrow cheaply in its own currency.

Challenges and Risks of Reserve Status

- **For Holding Countries**

- Vulnerability to the issuer's domestic policies (e.g., US sanctions or inflation).

- **Triffin Dilemma (Robert Triffin, 1960)**

- Conflict between domestic monetary policy and global liquidity needs.
- Over-issuance erodes confidence in the reserve asset's value.

- **Global Imbalances**

- Persistent US deficits (funded by reserve status) can lead to financial crises (e.g., the 2008 crisis).

- **Dedollarization Efforts**

- Countries (China, Russia) promoting alternatives to reduce reliance on the USD's "weaponization" via sanctions.

Historical Evolution: The Cycle of Dominance

- **Pre-1450:** Byzantine solidus, Florentine florin, Venetian ducat.
- **1450-1530:** Portuguese real; **1530-1640:** Spanish real (silver-based empire).
- **1640-1720:** Dutch guilder (trade dominance).
- **1720-1815:** French Livre
- **1815-1920:** **British Pound Sterling** (Backed by Gold Standard and Empire's trade).
- **Post-WWII (1944):** Bretton Woods Agreement established the **US Dollar** (USD) as the primary reserve currency, pegged to gold.
- **1971:** Nixon Shock ended the gold peg, transitioning to a pure fiat reserve system.
- **Modern Era:** USD dominance continues, but with rising challengers (EUR, CNY).

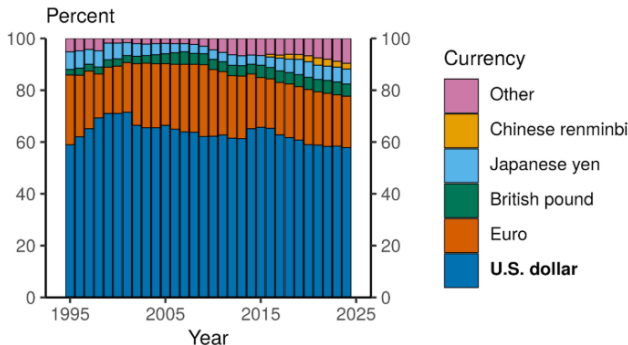
Observation: Average lifespan is ≈ 100 years. Transitions often follow the incumbent's economic decline and the rise of a new power.

Current Global Foreign Exchange Reserves

Global Holdings Remain Dominated by USD and EUR

Reserve currencies are held worldwide as safe, liquid assets.

- **USD Dominance:** Dominance persists despite a decline from over 70% in the early 2000s.
- **Renminbi (CNY):** Limited by capital controls and credibility concerns, restricting its growth.



Current Status of Reserve Currencies

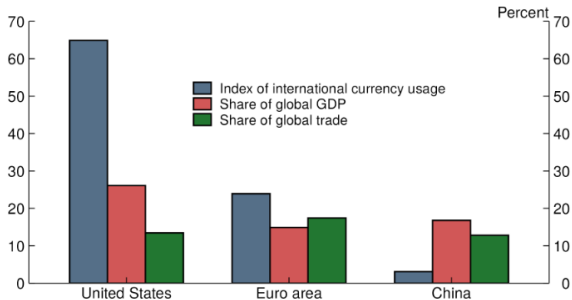
Based on IMF COFER data (as of Q2 2025):

Currency	Share (%)
US Dollar (USD)	58
Euro (EUR)	20
Japanese Yen (JPY)	5.5
British Pound (GBP)	4.8
Chinese Yuan (CNY)	2.8
Canadian Dollar (CAD)	2.5
Australian Dollar (AUD)	2.0
Others	4.4

Table: Approximate Global Reserve Shares (Q2 2025)

Note: USD share has declined from over 70% in the 2000s due to diversification and geopolitical shifts.

International Role vs. Size of Economy (2024)



Source:

<https://www.federalreserve.gov/econres/notes/feds-notes/the-international-role-of-the-u-s-dollar-2025-edition-20250718.html>

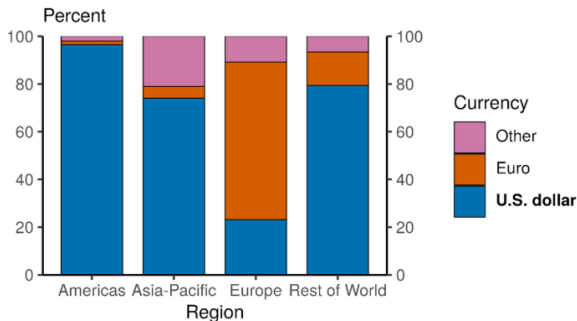
The US Dollar's Multifaceted Dominance

Why the USD remains the 'World's Currency'

- **Reserve Share:** 58% of global reserves (2024).
- **Trade and FX:** $\approx 88\%$ of all FX trades involve USD (BIS 2022); 96% of Americas invoicing.
- **Payment Systems:** $\approx 50\%$ of global SWIFT payments.
- **The Core Support:**
 - Deepest, most liquid **Treasury Market** (Safe Assets).
 - Support from **Fed Swap Lines** (critical backstop liquidity).
 - Strong institutional framework (**Rule of Law**).

Share of Export Invoicing

Usage as a medium of exchange and unit of account.



Source:

<https://www.federalreserve.gov/econres/notes/feds-notes/the-international-role-of-the-u-s-dollar-2025-edition-20250718.html>

Other Shares

Usage as a medium of exchange and unit of account.

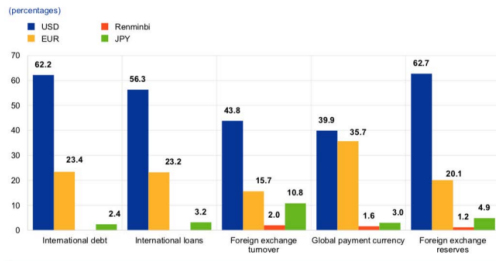


Figure 1: The Dominance of the dollar. *Source:* ECB Report on the international role of the euro, chart 2, [ECB \(2018\)](#).

Source: The International Monetary and Financial System

The Triffin Dilemma: The Core Instability

Conflict between Liquidity and Confidence (1960)

The Core Conflict

- **Liquidity Provision:** To supply the world with sufficient currency (L) for growing trade, the issuer (US) must run **Balance of Payments deficits**.
- **Confidence Requirement:** As foreign liabilities (L_{ext}) exceed the issuer's gold/asset backing (R_{us}), confidence in the currency's value **erodes**.

Instability $\implies$ Crisis of Confidence

$$\lim_{t \rightarrow \infty} \frac{L_{ext}}{R_{us}} \rightarrow \infty$$

Strategic Implications: Privilege and Constraint

Exorbitant Privilege

- **Seigniorage:** An interest-free loan from the rest of the world.
- **Yield Differential:** US earns higher returns on foreign assets (FDI) than it pays on liabilities (Treasuries).
- The US acts as the "World Venture Capitalist" (Gourinchas & Rey, 2007).

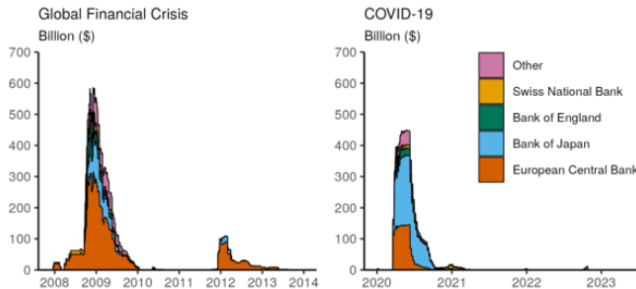
Original Sin

- The inability of developing nations to borrow abroad in their own currency.
- Leads to **Balance Sheet Mismatch**: Assets in local currency, Liabilities in FX.
- Currency depreciations become contractionary due to foreign debt servicing.

Strategic Implications: Privilege and Constraint 2

- **Policy Spillovers:** Fed policy drives the Global Financial Cycle (Rey, 2015). Fed swap lines provide critical backstop liquidity (e.g., \$450bn in 2020).
- **Imbalances:** Global Imbalances contribute to crises via "Excess Elasticity" rather than "Global Saving Glut." European banks were key drivers of the US boom.
- Gross flows exceed nets: Enable credit creation without deficits, challenging Triffin dilemma.
BIS data: Gross flows $>30\%$ world GDP pre-GFC.

FED swap lines



Source:

<https://www.federalreserve.gov/econres/notes/feds-notes/the-international-role-of-the-u-s-dollar-2025-edition-20250718.html>

Strategic Implications: Geopolitics and Policy

- **Global Financial Cycle (Hlne Rey, 2015):**

- Fed policy (interest rates) drives global credit and capital flows, limiting monetary autonomy for many nations.
- **Fed Swap Lines** provide critical USD liquidity backstops to central banks.

- **Weaponization and Fragmentation:**

- Use of sanctions "weaponizes" the USD (e.g., Russia 2022), fueling dedollarization rhetoric.
- Alternatives remain fragmented (e.g., CIPS vs. SWIFT).

- **The Future:** CBDCs may bypass the USD's vehicle currency role in cross-border settlement; **The New Triffin Dilemma:** Can the US supply safe assets (*T – Bills*) indefinitely without debt sustainability concerns?

"The dollar is our currency, but it's your problem." – John Connally (1971)

OCA: Definition and Core Theory

Definition

An OCA is a region where adopting a single currency maximizes economic benefits by outweighing the costs of losing independent monetary policy.

Robert Mundell (1961): The Theory of Factor Mobility

- **Problem:** Two regions (A and B) share a currency. A suffers a demand shock, B booms. A cannot devalue. (Focus on asymmetric shocks and adjustment mechanisms.)
- **Mechanism 1: Price/Wage Flexibility.** If W_A falls relative to W_B , equilibrium restores. (Often sticky downwards).
- **Mechanism 2: Labor Mobility.** If labor migrates $A \rightarrow B$, unemployment in A falls, inflationary pressure in B cools.

Mundell's Criterion

An OCA exists where **factor mobility** is high enough to substitute for exchange rate flexibility in the face of **asymmetric shocks**.

Key Criteria for Optimum Currency Areas

What is needed to manage Asymmetric Shocks?

- **Mundell (1961):** High **Labor Mobility** (Migration of workers).
- **McKinnon (1963):** High **Openness to Trade** (Exchange rates become ineffective).
- **Kenen (1969):** **Product Diversification** (Reduces vulnerability to sector-specific shocks).
- **Macro/Political:**
 - **Fiscal Integration and Transfers:** Centralized risk-sharing mechanism for redistribution during shocks. (**Fiscal Federalism**).
 - **Similar Business Cycles:** Synchronized economic fluctuations.
 - **Financial Integration:** Shared banking and capital markets for risk-sharing.
- **Price and Wage Flexibility:** Ability to adjust wages and prices downward in response to shocks, avoiding devaluation needs. (In practice, there is not true price/wage flexibility.)

Extensions to OCA Theory

The literature expanded the criteria for optimality beyond labor mobility:

- **McKinnon (1963) - Openness:** Highly open economies benefit more from fixed rates (price stability) than closed ones. The pass-through of exchange rates to domestic prices is higher in open economies, rendering devaluation ineffective.
- **Kenen (1969) - Product Diversification:** Diversified economies are less prone to asymmetric terms-of-trade shocks. Specialized economies need independent currencies as shock absorbers.
- **Fiscal Federalism (The "Transfer Union"):** Mechanism to redistribute income from booming to recessing regions (e.g., US Federal Tax system vs. EU budget).

Endogeneity of OCA Criteria

Can a Currency Union Create an OCA?

Is an OCA a static property or a dynamic process?

- **Traditional View (Krugman):** Specialization leads to asymmetric shocks. Integration makes OCA *less* likely.
- **Endogeneity View (Frankel & Rose, 1998):** Currency union increases trade intensity $\rightarrow$ increases business cycle correlation.

$$\text{Currency Union} \implies \text{Trade} \uparrow \implies \text{Corr}(Y_i, Y_j) \uparrow$$

Implication: Countries may satisfy OCA criteria *ex-post* even if they fail *ex-ante*.

Criticisms of OCA Theory

- Underestimates benefits: Ignores credibility gains, endogeneity (integration deepens post-union).
- Lacks operational criteria: Hard to measure/quantify thresholds.
- Ignores political factors: EMU driven by politics, not pure economics.
- Transition economies: May not apply; fixed rates beneficial regardless.

Benefits and Costs of OCAs

Benefits ✓

- **Transaction Efficiency:** Eliminates currency conversion costs.
- **Price Transparency:** Fosters greater competition.
- **Monetary Stability:** Larger area resists speculative attacks.
- **Trade Boost:** Eurozone trade increased 5-15% post-Euro.
- **Political Integration** Often accompanies deeper economic and political ties (e.g., EU).

Costs ✕

- **Loss of Autonomy:** Cannot devalue or adjust interest rates independently.
- **Asymmetric Shocks:** Regions hit differently with no adjustment tool (e.g., housing bubble in Spain vs. Germany in Eurozone).
- **Fiscal Spillover:** Debt crises can spread without strong rules (e.g., Greek debt crisis 2010-2015).
- **Political Tensions:** Resentment over bailouts and transfers.

Monetary Unions

- A **Monetary Union** is a rigid exchange rate arrangement where members share a common currency and a single central bank.
- Distinction: It differs from "Dollarization" because all members (theoretically) impact the common monetary policy.
- **Risk:** If member countries do not fulfill the conditions of an *Optimum Currency Area*, the union creates economic divergence.

Currency	Number of countries	Combined GDP, bil. USD, 2020	ERR
Euro	19	12933	Free floating
Western CFA franc	8	128	Pegged to euro
Central CFA franc	6	113	Pegged to euro
Eastern Caribbean dollar	8	5,5	Pegged to US dollar

Examples of Currency Areas

- **United States (Successful OCA):** High labor mobility and automatic federal transfers buffer asymmetric shocks (e.g., Texas oil bust vs. California tech boom).
- **Eurozone (Flawed OCA):**
 - Intended to become an OCA endogenously.
 - **Crisis (2010-2012):** Revealed low labor mobility and lack of Fiscal Union.
 - Resulted in "Internal Devaluation" (painful wage compression) rather than currency devaluation.
- **CFA Franc Zones:** Political peg to Euro/French Treasury; distinct dynamics.

Reserve Currencies and OCA: The Connection

- **Global OCA Anchor:** The USD acts as a de facto global anchor, reducing the immediate need for smaller regional currency unions.
- **Counterweight Motivation:** The Euro was explicitly designed not just for economic efficiency but also as a political counter-weight to US Dollar dominance ("Exorbitant Privilege").
- **Financial Spillovers:** The reserve issuer's (Fed's) policies drive the Global Financial Cycle, creating stability issues for dollar-linked emerging markets.
- **Risk Transfer:** Instability in a major OCA (e.g., the Eurozone Crisis) can undermine global reserve confidence, transferring systemic risk.

Case Study: The Eurozone (Flawed OCA)

The 2010-2012 Sovereign Debt Crisis

- ➊ **Initial Flaw:** Launched without critical OCA criteria (specifically, a full **Fiscal Union**).
- ➋ **Asymmetric Shocks:** The 2008 crisis hit the Periphery (Greece, Spain, Italy) much harder than the Core (Germany).
- ➌ **Adjustment Failure:** With no ability to devalue the currency, the only option was painful "**Internal Devaluation**" (wage compression and deflation).
- ➍ **Capital Flight (Sudden Stop):** Capital fled Periphery for Core (TARGET2 imbalances).
- ➎ **The Doom Loop:** Sovereign debt downgrades crippled banks holding that debt, requiring state bailouts, which further increased sovereign debt.

Resolution: ECB OMT ("Whatever it takes") + subsequent reforms (Banking Union, ESM).

Case Studies: Successful and Challenged OCAs

United States

High labor mobility, fiscal union, diversified economy – meets most criteria effectively.

Other Examples

Eastern Caribbean Currency Union (ECCU): Small islands sharing ECU, stable but limited scope. Proposed: African Monetary Union or Asian Currency Unit – face hurdles in integration.

Conclusion

- ① **Dominance:** Reserve currencies, led by USD, underpin international finance through liquidity and stability. Dominance persists despite the rise of multipolarity rhetoric.
- ② **Theoretical Basis:** OCA theory guides the formation of monetary unions, highlighting the trade-off between transaction efficiency and macroeconomic stability.
- ③ **The Dangers:** The Eurozone crisis demonstrated the dangers of forming a monetary union without meeting OCA criteria (specifically fiscal integration).
- ④ **Possible Future:** The system is evolving with digital assets (CBDCs) and fragmentation, but network effects keep the incumbent (USD) strong.

Sources

-  Mundell, R. A. (1961). A Theory of Optimum Currency Areas. *American Economic Review*.
-  Kenen, P. (1969). The Theory of Optimum Financial Areas.
-  Gourinchas, P. O., Rey, H., Sauzet, M. (2019). The international monetary and financial system.
-  Cohen, B. J. (1971). The Future of Sterling as an International Currency.
-  Frankel, J. A., & Rose, A. K. (1998). The Endogeneity of the Optimum Currency Area Criteria. *The Economic Journal*.
-  Farhi, E., & Maggiori, M. (2018). A Model of the International Monetary System. *QJE*.
-  Rey, H. (2015). Dilemma not Trilemma: The Global Financial Cycle and Monetary Policy Independence.
-  Federal Reserve Board (2025). The International Role of the U.S. Dollar.